Experiment: 10

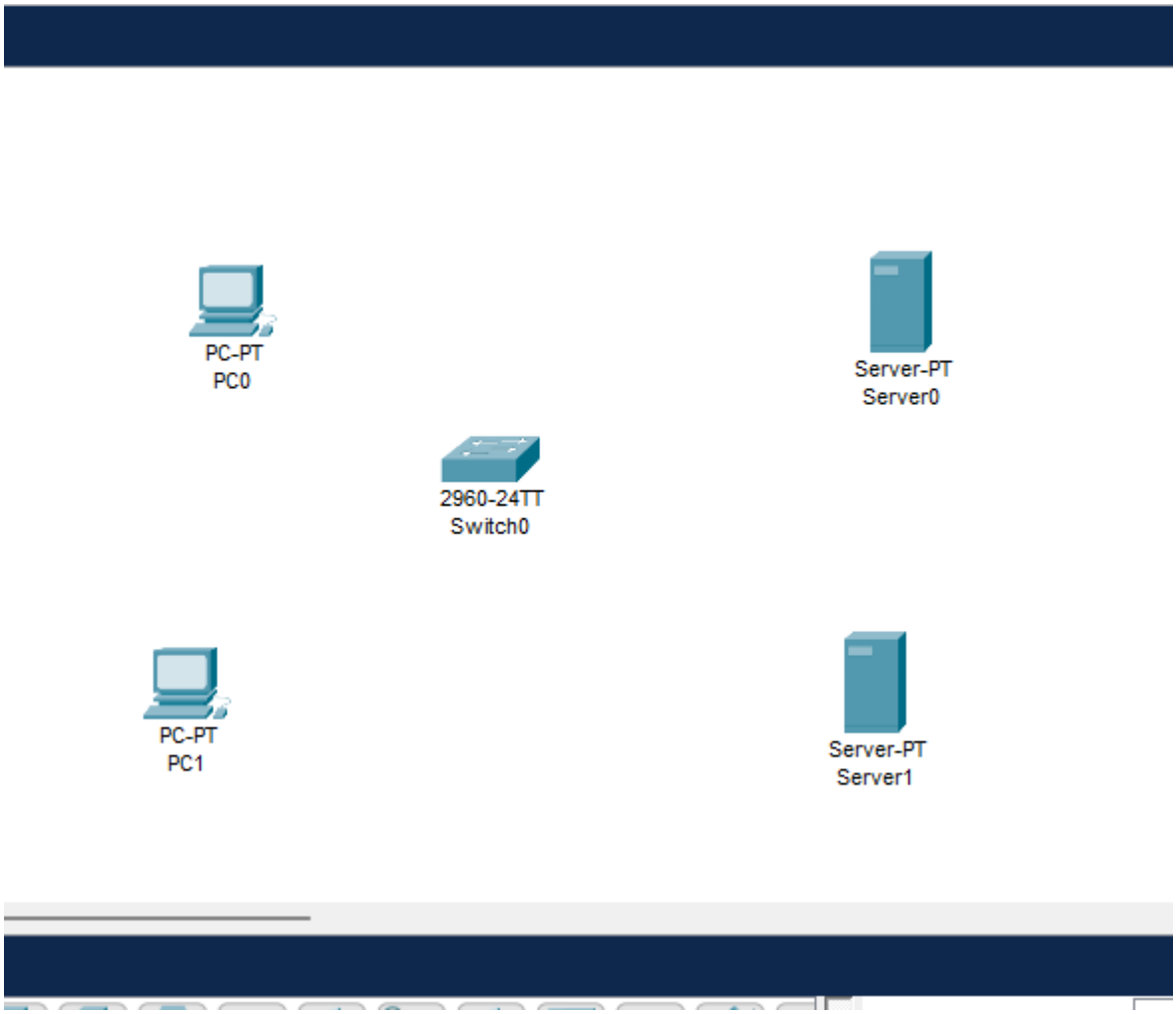
Problem Statement:-

Implement a DNS (Domain Name System) and DDNS (Dynamic Domain Name System) simulation to demonstrate domain name resolution and dynamic updates. Create a simple client-server application within Cisco Packet Tracer that queries the DNS for domain names and updates DNS records dynamically.

Aim:- DNS and DDNS Simulation: Domain Name Resolution and Dynamic Updates

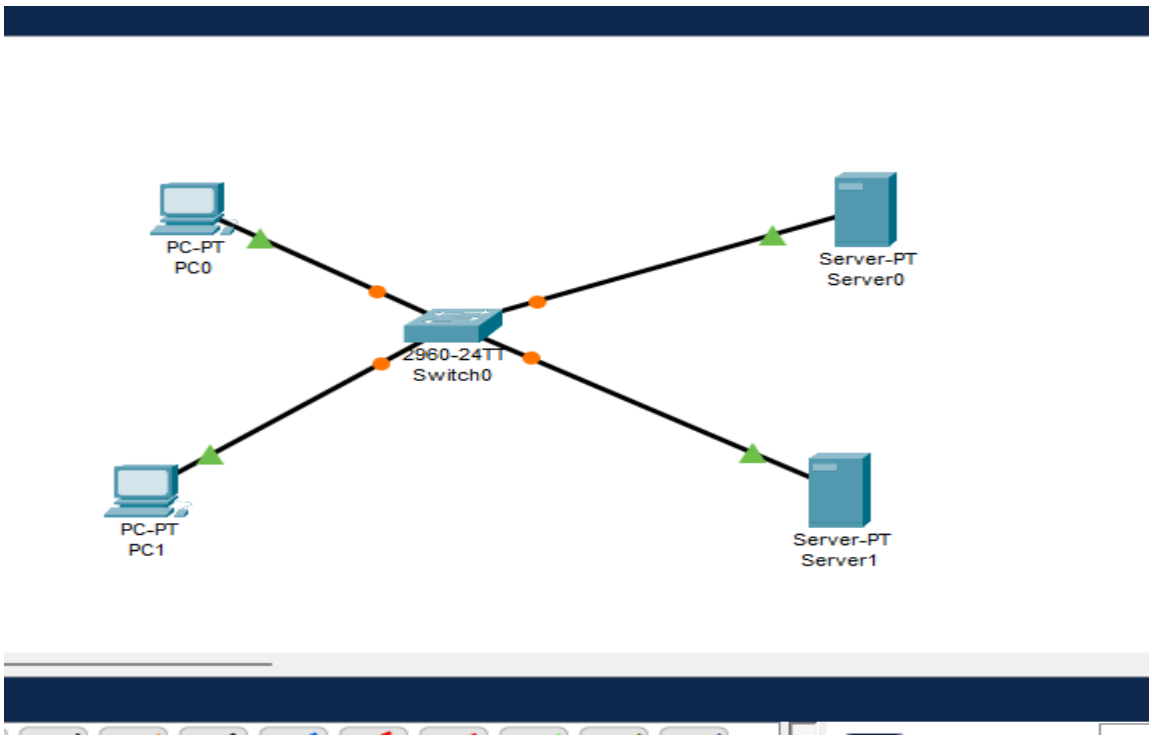
Procedure:-

Step 1: Add Devices to Workspace

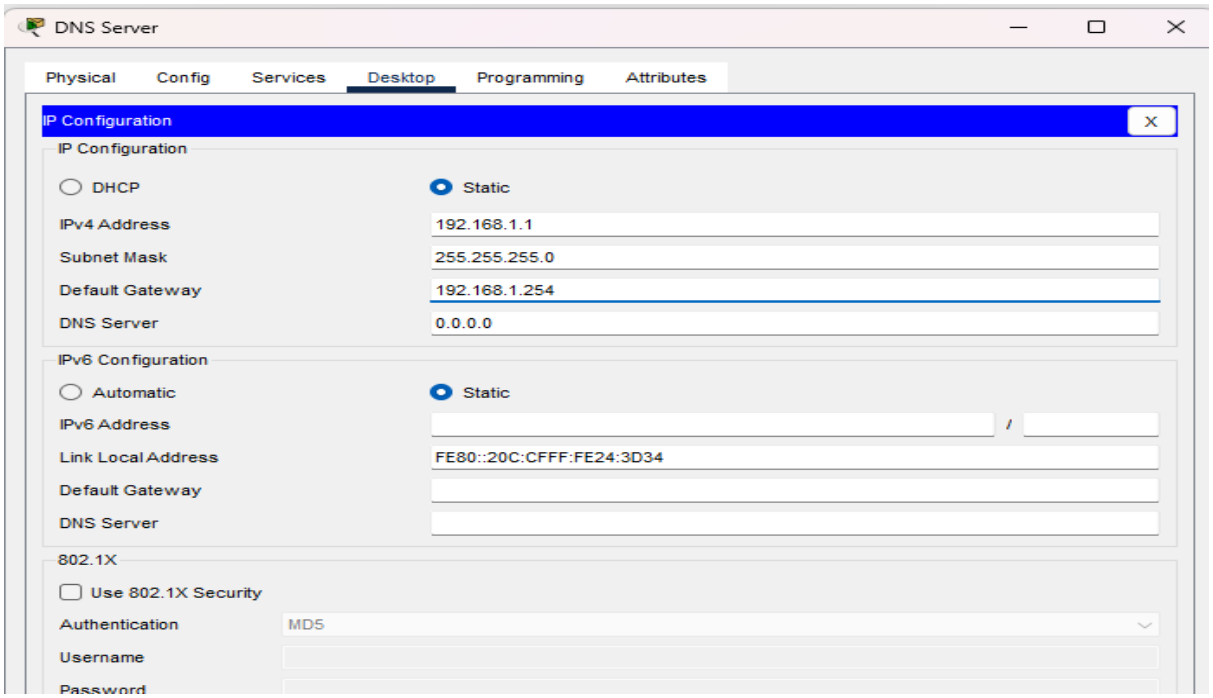


Step 2: Connect All Devices

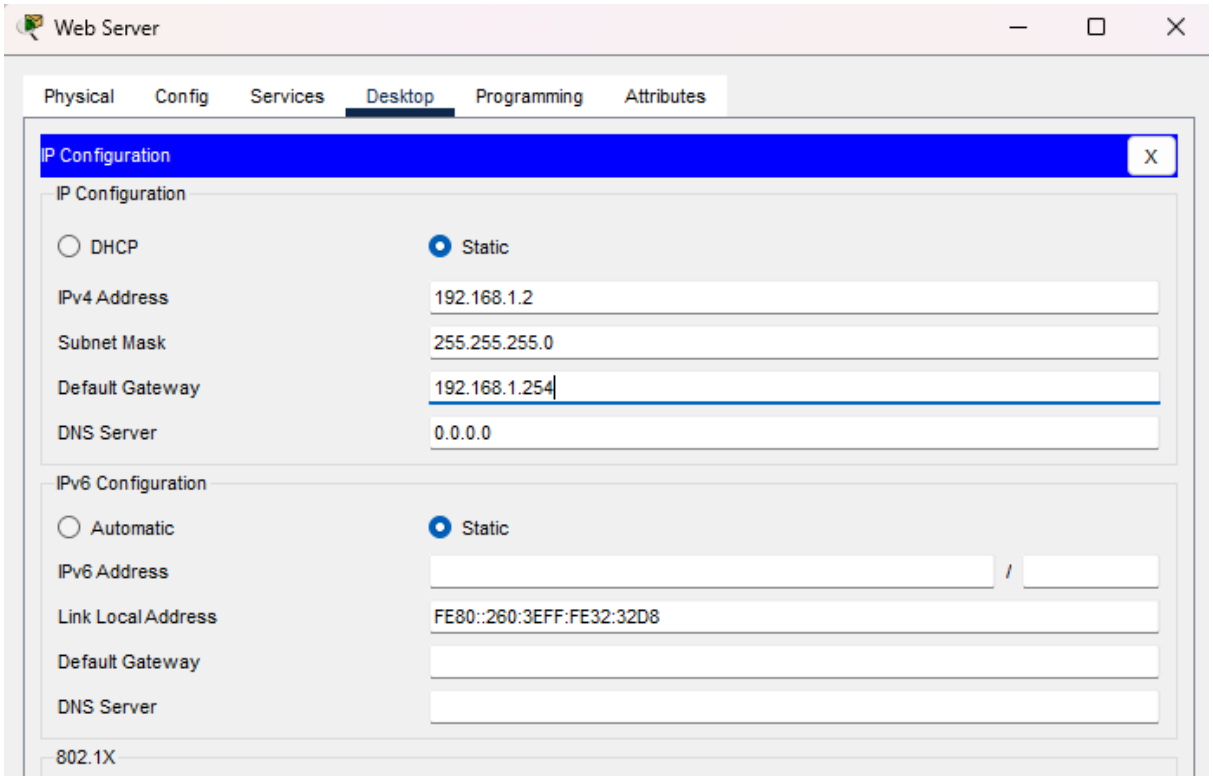
Use **Straight-Through Copper Cable:**



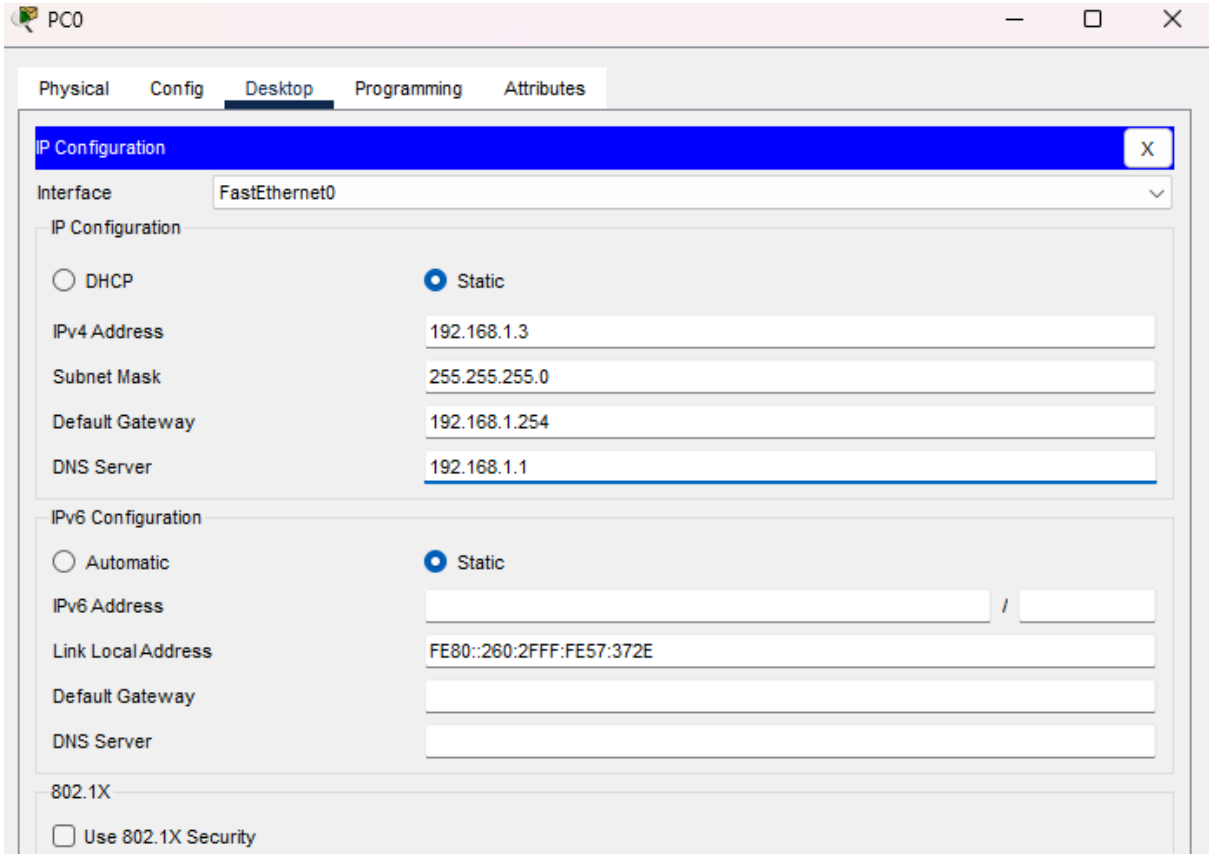
Step 3: Assign IP to DNS Server

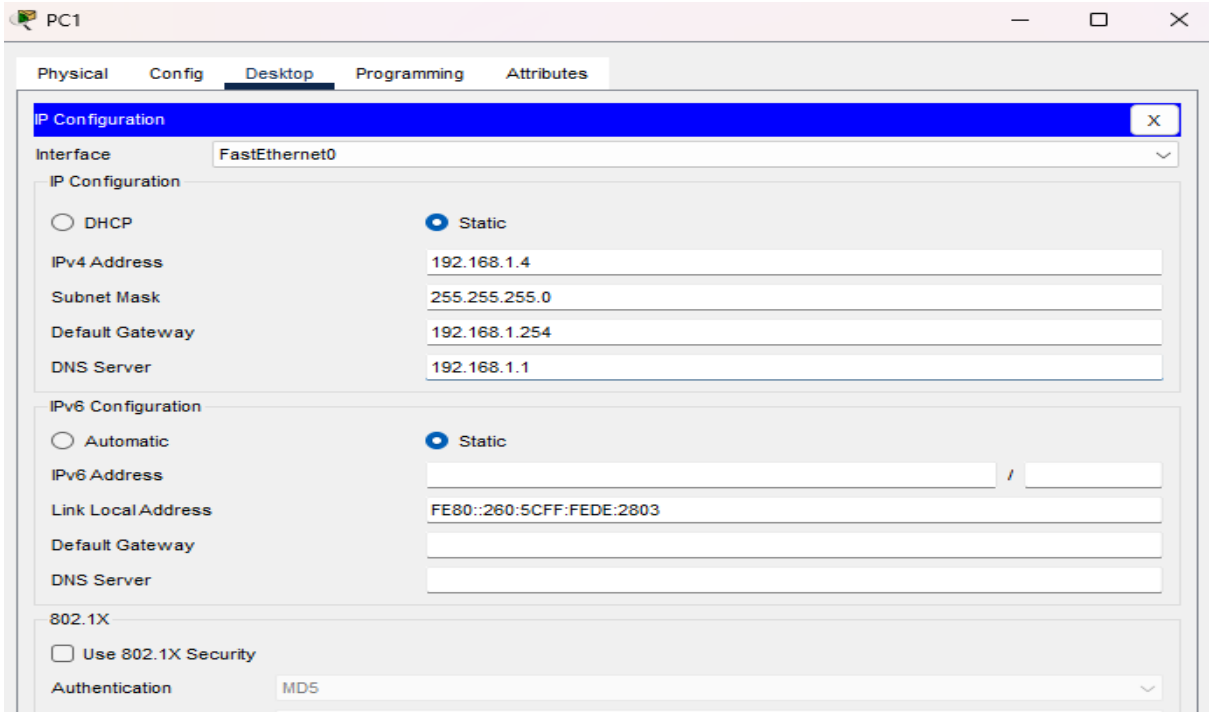


Step 4: Assign IP to Web Server

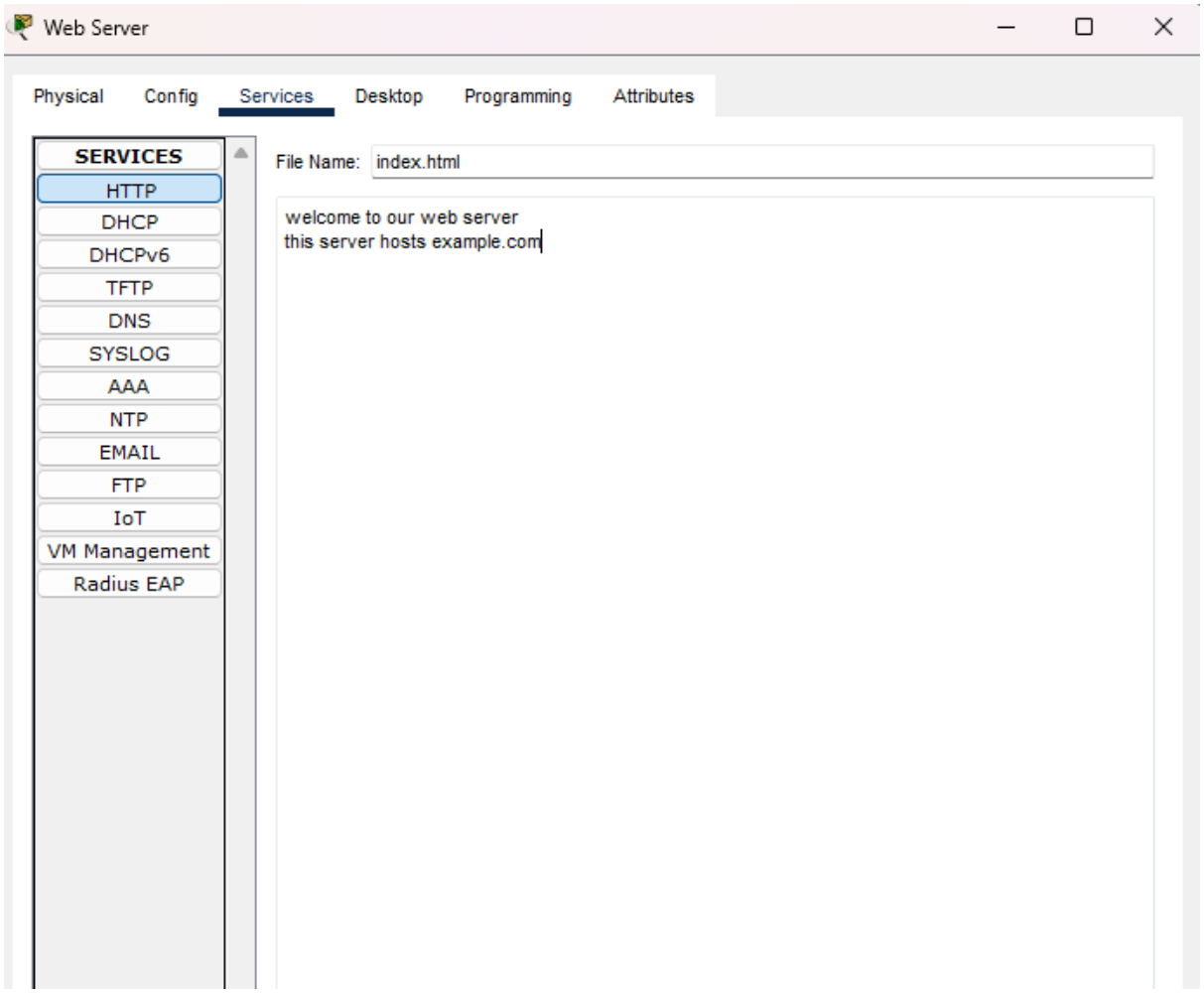


Step 5: Assign IP to PCs

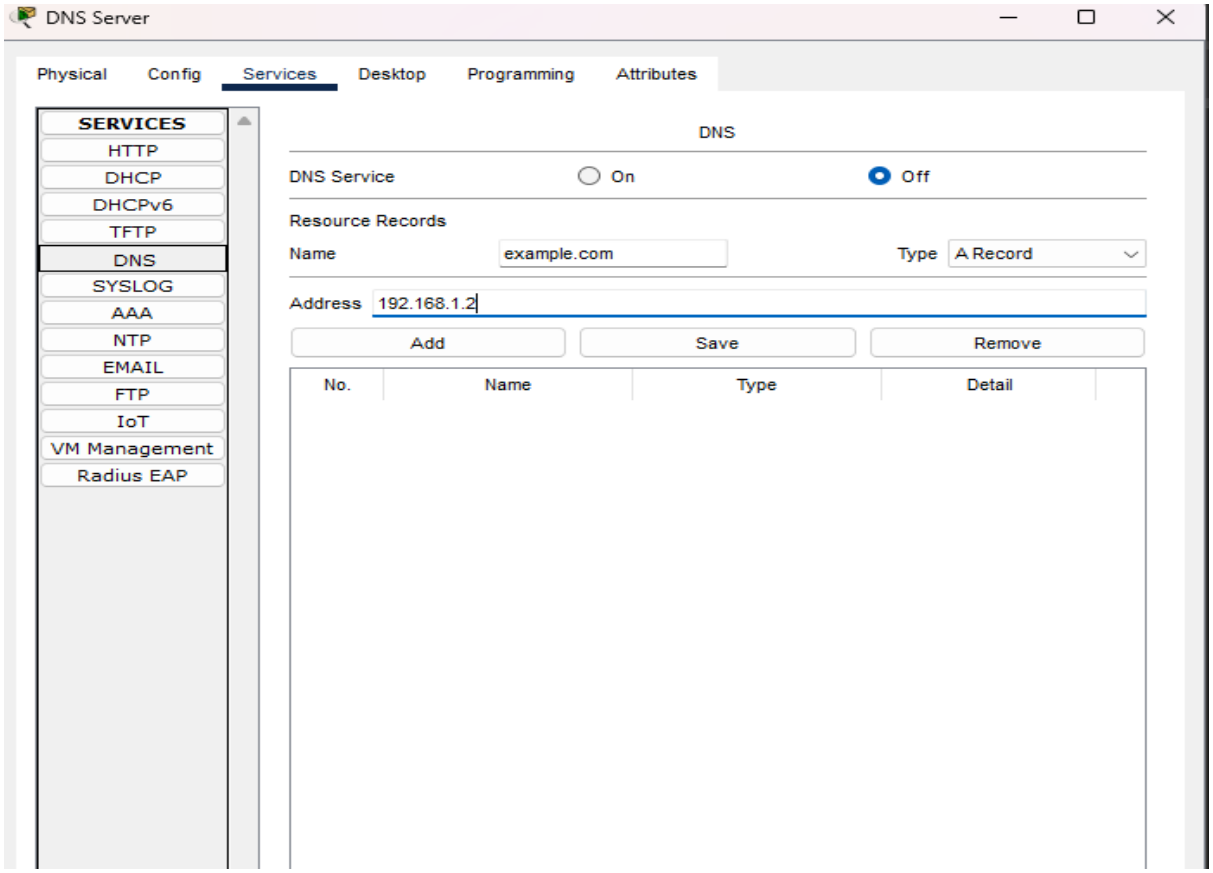


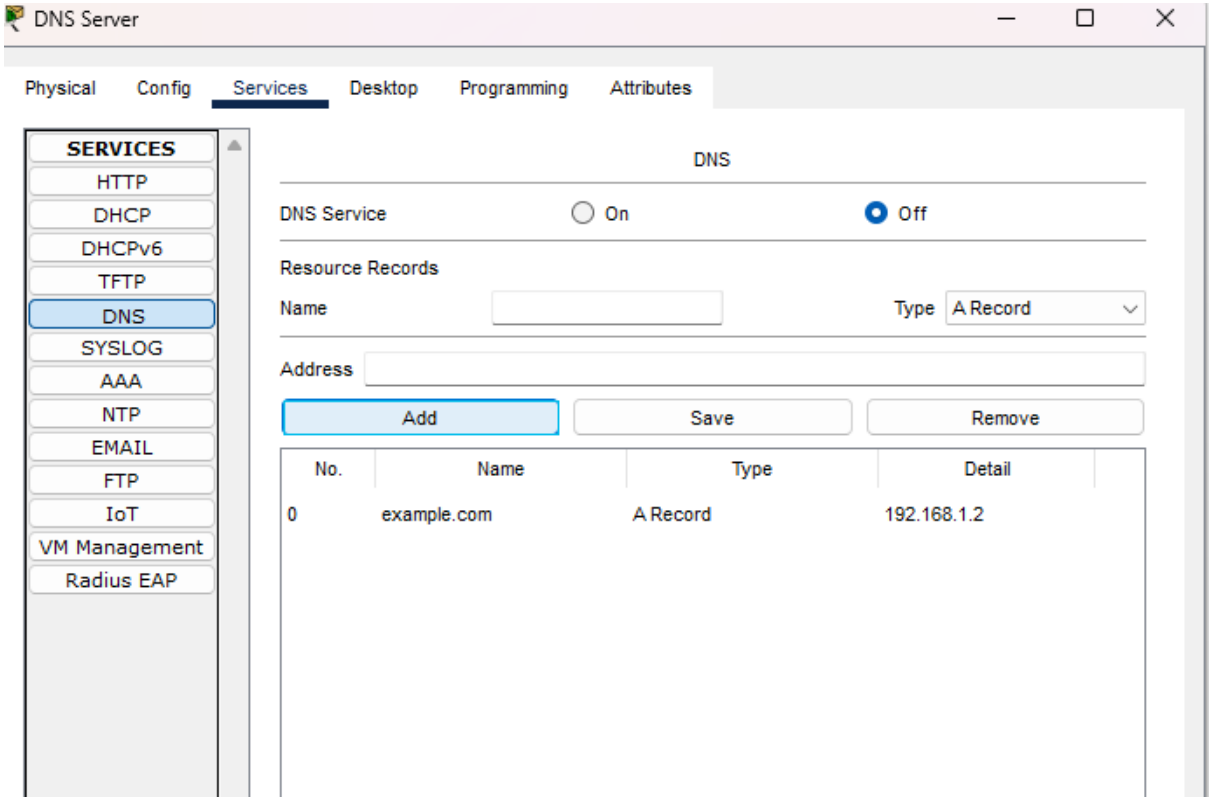


Step 7: Enable HTTP on Web Server



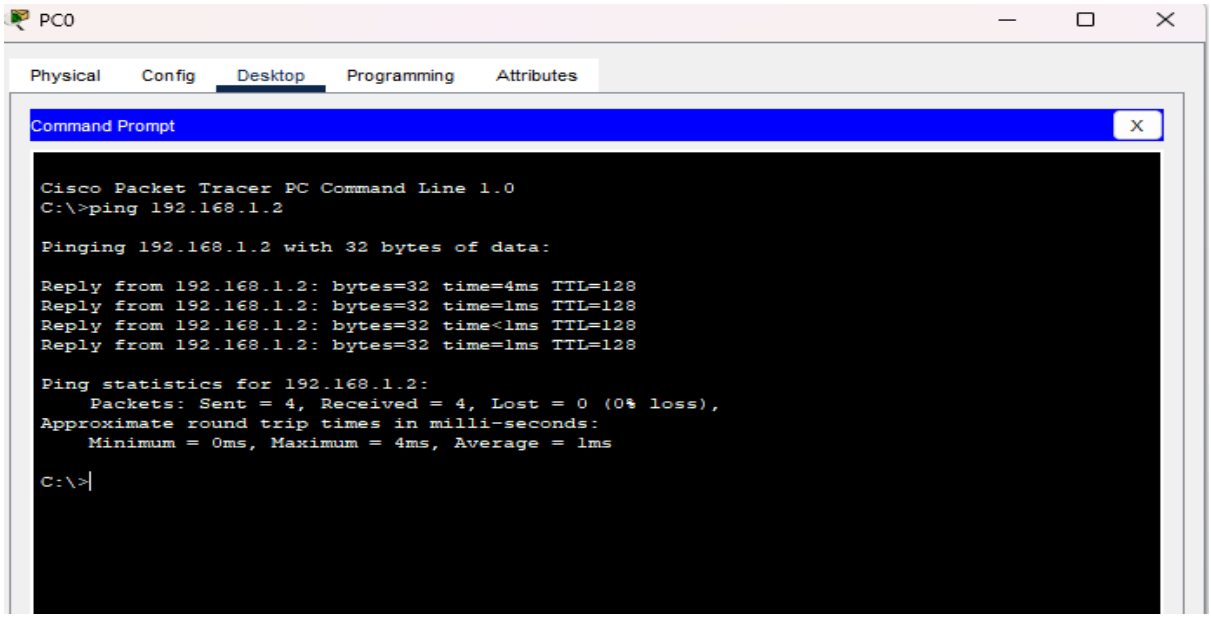
Step 8: Enable DNS Service





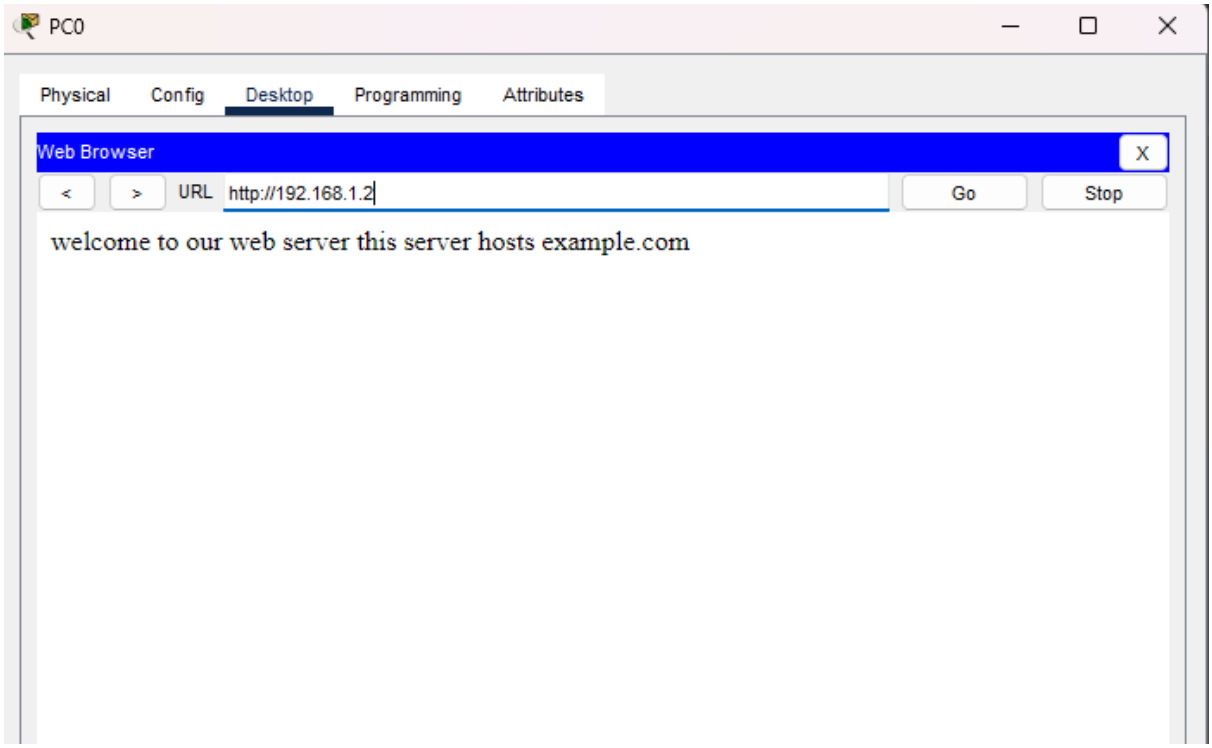
Step 10: Ping Test from PC0

1. Double-click **PC0**
2. Go to **Desktop → Command Prompt**
3. Type:



Step 11: Test DNS via Web Browser on PC0

1. Double-click **PC0**
2. Go to **Desktop → Web Browser**
3. Type in address bar:



Step 12: Test DNS via nslookup on PC1

1. Double-click **PC1**
2. Go to **Desktop → Command Prompt**
3. Type:

```
*** Unknown can't find test.com: Non-existent domain.

C:\>nslookup example.com

Server: [192.168.1.1]
Address:  192.168.1.1

Non-authoritative answer:
Name:   example.com
Address:   192.168.1.2

C:\>
```